KIMATHI UNIVERSITY COLLEGE OF TECHNOLOGY
UNIVERSITY EXAMINATIONS 2011/2012

SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
MECHANICAL ENGINEERING

EMG 2206: ENGINEERING THERMODYNAMICS 1

DATE: 21ST DECEMBER 2012

TIME: 2.00PM – 4.00PM

INSTRUCTIONS

Attempt any THREE QUESTIONS
All questions carry equal 20 marks each
Be neat and tidy
Use clear diagrams where necessary
You may assume air to be perfect gas $R = 0.287 KJ/KgK$, $\gamma = 1.4$

Q.1 (a) Give a concise definition of each of the following terms as used in Thermodynamics:
(i) System
(ii) Property
(iii) State
(iv) Process
(v) Thermodynamic Cycle (5 mrks)

(b) Give two similarities between heat and work (2 mrks)

(c) (i) The pressure in a piston – cylinder arrangement varies with volume according to $PV^2 = C$, where C is constant. The initial pressure is 5 bar and the initial volume is $0.05m^3$, and the final pressure is 2 bar. Find the work done. (7 mrks)

(ii) A room 5m x 4m contains 72kg of air at a pressure of 1.05 bar. Determine the temperature of the air in the room. (6 mrks)

Q.2 (a) Define Reversibility (2 mrks)

(b) Write the steady flow energy equation
(i) for unit mass (1 mrk)
(ii) for unit weight (1 mrk)

Also, write the continuity of mass equation. (1 mrk)

(c) Air flows steadily at the rate of 0.4kg/s through an air compressor, entering at 5m/s with a pressure of 1 bar and specific volume of $0.85m^3/Kg$, and leaving at 4.5m/s with a pressure of 6.9 bar and specific volume of $0.16m^3/Kg$. The specific internal energy of the air leaving is 88 KJ/Kg greater than that of air entering. Cooling water in the jacket surrounding the clylinder absorbs heat from the air at the rate of 59 KJ/s. Calculate:

(i) The power required to drive the compressor (5 mrks)
(ii) The inlet pipe cross-sectional area (5 mrks)
(iii) The outlet pipe cross-sectional area (5 mrks)

Q.3 (a) Define reversible work (1 mrk)

(b) Determine the work done by a gas during expansion between two points for a polytropic process $PV^n = C$.

(4 mrks)

(c) A quantity of air has a volume of $0.15 m^3$ at a temperature of $120^0 C$ and a pressure of 12 bar. The air expands to a pressure of 2 bar according to the law $PV^{1.32} = C$

Determine:

(i) The work transfer (5 mrks)
(ii) The change in internal energy (5 mrks)
(iii) The heat transfer (5 mrks)

Take $C_p = 1.006$ KJ/KgK
$C_v = 0717$ KJ/KgK

Q.4 (a) State the following as used for perfect gas:-
(i) Boyle's law
(ii) Charles Law
(iii) Avogadro's Hypothesis

(3 mrks)

(b) Using the Non-flow equation and polytropic expression show that

$$Q = \left[\frac{\gamma - n}{n-1}\right] W$$

(usual notation used) (5 mrks)

(c) One kilogramme of a perfect gas is compressed from 1.1 bar, $27^0 C$ until the pressure is 6.6 bar according to the law $Pv^{1.3}$ = constant.

Calculate, the heat flow, to or from the cylinder walls:

(i) When the gas is Ethane (molar mass 30Kg/Kmol) with $C_p = 2.10$ KJ/KgK

(6 mrks)

(ii) When the gas is Argon (molecular mass 40 Kg/Kmol) with $C_p = 0.520$KJ/KgK

(6 mrks)

Q.5. (a) Write down the Non-flow energy equation and state the sign convention used with this equation.

(2 mrks)

(b) i) Sketch the following processes for a gas in each case indicating the index of the Process;

(3 mrks)

ii) For each of the following reversible processes write down the non-flow energy equation and indicate how each of the terms in the equation could be evaluated for <u>both</u> gases and vapours.

(i) Isothermal compression
(ii) Adiabatic expansion
(iii) Isobaric Cooling
(iv) Isochoric Heating
(v) Polytropic Compression

(5 mrks)

(c) Complete the following table showing clearly your working and sketches on a PV diagram:

	Fluid	Pressure p (bars)	Temperature t ^{0}C	Dryness Fraction x	Specific Volume v (m^3/kg)	Specific Internal u KJ/kg	Specific Enthalpy h (KJ/kg)	Specific Entropy s (KJ/kgK)
1	H_20		230	0.6				
2.	NH_3		-10	Saturated Vapour				
3.	Freon-12	2.61	5					
4.	NH_3	6.585	30					
5.	Freon-12	0.168	-45					

Assume for superheated steam

$$v_{sup} = \frac{v_g \times T\ sup\ (K)}{T\ sat\ (K)}$$

(10 mrks